Problem · Unit 4 · 10 points · 60 min · High difficulty / trap

P02 — Particle on a smooth paraboloid of revolution: axial angular momentum, circular orbits and radial oscillations

*A particle P of mass m moves on the **inner** face of a smooth, fixed paraboloid of revolution whose axis Oz is vertical (upward; uniform gravity $-g\,\mathbf{k}$). In cylindrical coordinates (r, φ, z) of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ the surface is*

$$z = \frac{r^2}{2c} \qquad (c > 0)$$

with O at its vertex. The particle rests on the surface from above (the contact can only push P away from the surface, towards the inside of the bowl). The azimuth φ is measured from the x axis, counter-clockwise seen from above.

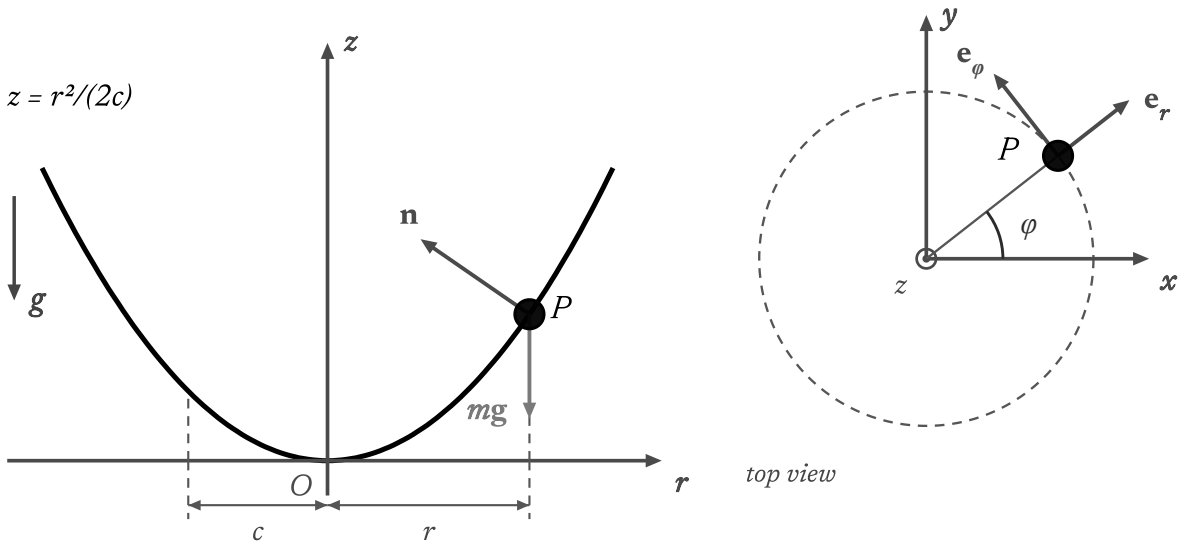


Figure 1. Particle P on the smooth inner surface $z = r^2/(2c)$ (axis Oz vertical, gravity $-g\,\mathbf{k}$). Left: meridian plane through P with the unit normal $\mathbf{n}$; right: top view with the azimuth φ.

Use the cylindrical basis $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}\}$, and write $q = \sqrt{1 + r^2/c^2}$ and $\Omega_0 = \sqrt{g/c}$.

Setup

- Data: m, c, g. In part (f): initial radius $r_0 = c/2$, $\dot r(0) = 0$, initial velocity horizontal and tangent to the parallel with speed $v_0 = 2r_0\Omega_0 = c\,\Omega_0$; for numbers, $c = 1$ m and $g = 9.81$ m s^{-2}.
- Hypotheses: point particle, no friction, S_0 inertial. The contact is **unilateral**: the model is valid while the normal reaction is non-negative (checked in (f)).

Part (a) · 1.5 points

Degrees of freedom and kinematics

How many degrees of freedom has P while it stays on the surface? Write its velocity and acceleration in the cylindrical basis using the constraint.

Step 1. Count

A free particle needs (r, φ, z). The surface $f = z - r^2/(2c) = 0$ is one holonomic equation without explicit time: $3 - 1 = 2$ degrees of freedom (Notes §2.7), with generalized coordinates r and φ. The constraint is unilateral, because the particle is only pressed against the surface.

Step 2. Derivatives of z

By the chain rule, $\dot z = \dfrac{dz}{dr}\dot r = \dfrac{r\dot r}{c}$. By the product rule, $\ddot z = \dfrac{1}{c}\dfrac{d}{dt}(r\dot r) = \dfrac{\dot r\,\dot r + r\ddot r}{c} = \dfrac{\dot r^2 + r\ddot r}{c}$.

Step 3. Change of basis

The cylindrical basis turns with φ; its relation to the inertial basis is

$$\left(\mathbf{e}_r\ \mathbf{e}_\varphi\ \mathbf{k}\right) = [R]\left(\mathbf{i}\ \mathbf{j}\ \mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: rows 1 and 2 have length $\cos^2\varphi + \sin^2\varphi = 1$ and $-\cos\varphi\sin\varphi + \sin\varphi\cos\varphi = 0$, row 3 is $\mathbf{k}$, so $[R][R]^T = I$; $\det[R] = \cos^2\varphi + \sin^2\varphi = +1$ (right-handed, $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{k}$). By the chain rule, $\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r$.

Step 4. Velocity and acceleration

In cylindrical coordinates (Notes §2.6.2), $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi + \dot z\,\mathbf{k}$ and $\mathbf{a} = (\ddot r - r\dot\varphi^2)\mathbf{e}_r + (r\ddot\varphi + 2\dot r\dot\varphi)\mathbf{e}_\varphi + \ddot z\,\mathbf{k}$. With Step 2:

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\varphi\,\mathbf{e}_\varphi + \frac{r\dot r}{c}\,\mathbf{k}, \qquad |\mathbf{v}|^2 = q^2\dot r^2 + r^2\dot\varphi^2$$

$$\mathbf{a} = (\ddot r - r\dot\varphi^2)\,\mathbf{e}_r + (2\dot r\dot\varphi + r\ddot\varphi)\,\mathbf{e}_\varphi + \frac{\dot r^2 + r\ddot r}{c}\,\mathbf{k}$$

KEY RESULTS

- 2 degrees of freedom, coordinates (r, φ); one holonomic, scleronomic, unilateral constraint.
- $\dot z = r\dot r/c,\ \ddot z = (\dot r^2 + r\ddot r)/c$.
- $\mathbf{a} = (\ddot r - r\dot\varphi^2)\mathbf{e}_r + (2\dot r\dot\varphi + r\ddot\varphi)\mathbf{e}_\varphi + \dfrac{\dot r^2 + r\ddot r}{c}\mathbf{k}$.

Part (b) · 1 point

Forces

Obtain the unit normal of the surface and write the forces on P.

Step 1. Normal vector

$\nabla f = \dfrac{\partial f}{\partial r}\mathbf{e}_r + \dfrac{1}{r}\dfrac{\partial f}{\partial \varphi}\mathbf{e}_\varphi + \dfrac{\partial f}{\partial z}\mathbf{k} = -\dfrac{r}{c}\,\mathbf{e}_r + \mathbf{k}$, of modulus $\sqrt{1 + r^2/c^2} = q$ (Notes §6.3). Its $\mathbf{k}$ component is positive, so it points into the bowl, the side on which the particle lies:

$$\mathbf{n} = \frac{1}{q}\Big(-\frac{r}{c}\,\mathbf{e}_r + \mathbf{k}\Big)$$

Check: $\mathbf{n}\cdot\mathbf{v} = \dfrac{1}{q}\Big(-\dfrac{r\dot r}{c} + \dfrac{r\dot r}{c}\Big) = 0$.

Step 2. Forces

The weight $\mathbf{W} = -mg\,\mathbf{k}$ and the smooth normal reaction $\mathbf{N} = N\,\mathbf{n}$ (Notes §6.4). Contact requires $N \geq 0$. Note that $\mathbf{N}$ lies in the meridian plane (it has no $\mathbf{e}_\varphi$ component) but does **not** pass through O in general.

KEY RESULTS

- $\mathbf{n} = \dfrac{-(r/c)\,\mathbf{e}_r + \mathbf{k}}{q}$ (towards the inside of the bowl).
- $\mathbf{W} = -mg\,\mathbf{k},\ \mathbf{N} = N\mathbf{n}$ with $N \geq 0$ for contact.

Part (c) · 2 points

Equations of motion and first integrals

Project Newton's law, show that the component $H = r^2\dot\varphi$ of the specific angular momentum along the axis is conserved, write the energy and obtain one equation for $r(t)$ free of N. Give N as a function of the state.

Step 1. Projections

With $\mathbf{a}$ of (a), $\mathbf{N}\cdot\mathbf{e}_r = -Nr/(cq)$, $\mathbf{N}\cdot\mathbf{k} = N/q$:

$$\mathbf{e}_r:\ m(\ddot r - r\dot\varphi^2) = -\frac{Nr}{cq} \qquad \mathbf{e}_\varphi:\ m(2\dot r\dot\varphi + r\ddot\varphi) = 0 \qquad \mathbf{k}:\ m\frac{\dot r^2 + r\ddot r}{c} = \frac{N}{q} - mg$$

Step 2. Axial angular momentum

By the product rule, $\dfrac{d}{dt}(r^2\dot\varphi) = 2r\dot r\dot\varphi + r^2\ddot\varphi = r(2\dot r\dot\varphi + r\ddot\varphi)$, so the $\mathbf{e}_\varphi$ equation is $\dfrac{m}{r}\dfrac{d}{dt}(r^2\dot\varphi) = 0$: $H = r^2\dot\varphi$ is constant. Physical reason (Notes §4.8): $\mathbf{k}\cdot(\mathbf{r}\times\mathbf{F})$ is the moment about the z axis; $\mathbf{W}$ is parallel to the axis and $\mathbf{N}$ lies in the meridian plane, which contains the axis, so both have zero moment about it. The full vector $\mathbf{H}_O$ is **not** conserved ($\mathbf{r}\times\mathbf{N} \neq \mathbf{0}$ in general): only its axial component is.

Step 3. Equation for r

From $\mathbf{k}$: $\dfrac{N}{q} = m\Big(\dfrac{\dot r^2 + r\ddot r}{c} + g\Big)$. Substituting in $\mathbf{e}_r$ (which contains $\dfrac{r}{c}\cdot\dfrac{N}{q}$) and using $\dot\varphi = H/r^2$:

$$\ddot r - \frac{H^2}{r^3} = -\frac{r}{c}\Big(\frac{\dot r^2 + r\ddot r}{c} + g\Big) \implies \Big(1 + \frac{r^2}{c^2}\Big)\ddot r + \frac{r\dot r^2}{c^2} - \frac{H^2}{r^3} + \frac{gr}{c} = 0$$

Step 4. Energy

$\mathbf{N}\cdot\mathbf{v} = 0$ (fixed smooth surface) and $\mathbf{W}$ derives from $V = mgz = mgr^2/(2c)$, so $E = \tfrac12 m|\mathbf{v}|^2 + V$ is constant (Notes §4.5). With $|\mathbf{v}|^2$ of (a) and $r\dot\varphi = H/r$:

$$\frac{E}{m} = \frac12\Big[\Big(1 + \frac{r^2}{c^2}\Big)\dot r^2 + \frac{H^2}{r^2}\Big] + \frac{gr^2}{2c}$$

Check: $\dfrac{d}{dt}\dfrac{E}{m} = \dot r\Big[\Big(1 + \dfrac{r^2}{c^2}\Big)\ddot r + \dfrac{r\dot r^2}{c^2} - \dfrac{H^2}{r^3} + \dfrac{gr}{c}\Big] = 0$ by Step 3 (the term $\dfrac{r\dot r^3}{c^2}$ comes from the chain rule on $\dfrac{r^2}{c^2}\dot r^2$).

Step 5. Reaction

$N = mq\Big(\dfrac{\dot r^2 + r\ddot r}{c} + g\Big)$, with $\ddot r$ from Step 3.

KEY RESULTS

- $H = r^2\dot\varphi = \text{const}$ (the moment about the axis of $\mathbf{W}$ and $\mathbf{N}$ is zero; $\mathbf{H}_O$ itself is not conserved).
- $\Big(1 + \dfrac{r^2}{c^2}\Big)\ddot r + \dfrac{r\dot r^2}{c^2} - \dfrac{H^2}{r^3} + \dfrac{gr}{c} = 0.$
- $\dfrac{E}{m} = \tfrac{1}{2}\Big[\Big(1 + \dfrac{r^2}{c^2}\Big)\dot r^2 + \dfrac{H^2}{r^2}\Big] + \dfrac{gr^2}{2c};\ N = mq\Big(\dfrac{\dot r^2 + r\ddot r}{c} + g\Big).$

Part (d) · 1.5 points

Horizontal circular orbits

Find the angular rate of a horizontal circular motion at radius r_0, its speed and the normal reaction. Discuss the dependence on r_0.

Step 1. Rate

With $\dot r = \ddot r = 0$ and $H = r_0^2\dot\varphi$: $\dfrac{H^2}{r_0^3} = r_0\dot\varphi^2 = \dfrac{gr_0}{c}$, hence

$$\dot\varphi^2 = \frac{g}{c} \quad\Longrightarrow\quad \Omega_0 = \sqrt{\frac{g}{c}}, \qquad T = 2\pi\sqrt{\frac{c}{g}}, \qquad v = r_0\Omega_0$$

All horizontal circles are described in the same time, whatever their radius (isochronous orbits). This is the trap of the item: on a cone or on a sphere the rate depends on the radius; on the paraboloid the slope $dz/dr = r/c$ grows exactly in proportion to the centripetal acceleration $r\Omega^2$ needed.

Step 2. Reaction

$N = mq_0(0 + g) = mg\sqrt{1 + r_0^2/c^2}$. Its vertical component $N/q_0 = mg$ carries the weight; its horizontal component $Nr_0/(cq_0) = mgr_0/c = mr_0\Omega_0^2$ is the centripetal force.

KEY RESULTS

- $\dot\varphi^2 = g/c$: $\Omega_0 = \sqrt{g/c}$, **the same for every** r_0; period $2\pi\sqrt{c/g}$; speed $r_0\Omega_0$.
- $N = mg\sqrt{1 + r_0^2/c^2}$.

Part (e) · 2 points

Small radial oscillations about a circular orbit

Perturb the circular orbit of radius r_0 keeping $H = r_0^2\Omega_0$. Obtain the frequency ω_r of the small radial oscillations and the ratio ω_r/Ω_0. For which r_0 is the perturbed orbit closed after one revolution? What happens for $r_0 \to 0$ and $r_0 = c$?

Step 1. Linearisation

$G(r,\dot r) = \dfrac{F(r) - r\dot r^2/c^2}{1 + r^2/c^2}$ with $F(r) = \dfrac{H^2}{r^3} - \dfrac{gr}{c}$, and $F(r_0) = 0$ by (d). Put $r = r_0 + \varepsilon$ with $\varepsilon, \dot\varepsilon$ small. $\partial G/\partial \dot r = -2r\dot r/(c^2 q^2)$ vanishes at $\dot r = 0$. Since $F(r_0) = 0$, $\left.\dfrac{\partial G}{\partial r}\right|_{r_0} = \dfrac{F'(r_0)}{1 + r_0^2/c^2}$ (the derivative of the denominator multiplies $F(r_0) = 0$). With $F'(r) = -\dfrac{3H^2}{r^4} - \dfrac{g}{c}$ and $H^2 = r_0^4\, g/c$:

$$F'(r_0) = -\frac{3g}{c} - \frac{g}{c} = -\frac{4g}{c} \qquad \Longrightarrow \qquad \ddot\varepsilon + \frac{4g}{c\,(1 + r_0^2/c^2)}\,\varepsilon = 0$$

Step 2. Frequency and ratio

$$\omega_r^2 = \frac{4gc}{c^2 + r_0^2}, \qquad \frac{\omega_r}{\Omega_0} = \sqrt{\frac{4c^2}{c^2 + r_0^2}} = \frac{2c}{\sqrt{c^2 + r_0^2}}$$

The orbit is stable (positive ω_r^2) for every r_0. The trap: the radial frequency is **not** Ω_0, and the factor $1 + r_0^2/c^2$ (the inertia of the vertical motion that comes with $\dot r$) cannot be dropped.

Step 3. Special radii

- $r_0 \to 0$: ratio 2: two radial oscillations per revolution, a centred ellipse, as for an isotropic harmonic bowl ($z \approx r^2/(2c)$ is then the potential of a linear spring of stiffness mg/c).
- $r_0 = c$: ratio $\sqrt{2}$, irrational: the perturbed orbit precesses and never closes.
- $r_0 = \sqrt{3}\,c$: ratio $2c/\sqrt{4c^2} = 1$: one radial oscillation per revolution, a closed orbit.

KEY RESULTS

- $\omega_r^2 = \dfrac{4g}{c\,(1 + r_0^2/c^2)} = \dfrac{4gc}{c^2 + r_0^2}, \quad \dfrac{\omega_r}{\Omega_0} = \dfrac{2c}{\sqrt{c^2 + r_0^2}}.$
- Ratio 1 for $r_0 = \sqrt{3}\,c$ (closed after one revolution); 2 for $r_0 \to 0$; $\sqrt{2}$ for $r_0 = c$ (not closed).

Part (f) · 2 points

Turning points and contact

The particle starts at r_0 with $\dot r = 0$ and a horizontal velocity v_0 tangent to the parallel. Find the turning radii. Then take $r_0 = c/2$, $v_0 = 2r_0\Omega_0$: give the turning radii, $\ddot r$ and N at both, and decide whether the particle ever leaves the surface.

Step 1. Constants

$\mathbf{v}(0) = v_0\mathbf{e}_\varphi$, so $H = r_0 \cdot r_0\dot\varphi(0) = r_0 v_0$ and $E/m = \tfrac12 v_0^2 + \dfrac{g r_0^2}{2c}$.

Step 2. Turning radii

At $\dot r = 0$ the energy of (c) gives $\dfrac{H^2}{r^2} + \dfrac{gr^2}{c} = v_0^2 + \dfrac{gr_0^2}{c}$. Multiply by $u = r^2$:

$$\frac{g}{c}u^2 - \Big(v_0^2 + \frac{gr_0^2}{c}\Big)u + r_0^2 v_0^2 = 0 \iff (u - r_0^2)\Big(\frac{gu}{c} - v_0^2\Big) = 0$$

(expanding the product gives back the three terms). Roots: $r = r_0$ and $r^* = v_0\sqrt{c/g} = v_0/\Omega_0$. If $v_0 > r_0\Omega_0$ the particle climbs to $r^* > r_0$; if $v_0 = r_0\Omega_0$ both coincide (circular orbit of (d)). The time between them is a quadrature: from the energy, $dt = \dfrac{q\,dr}{\sqrt{2E/m - H^2/r^2 - gr^2/c}}$.

Step 3. The numerical case

$r_0 = c/2$, $v_0 = c\Omega_0$: $r^* = c\Omega_0/\Omega_0 = c$. So r oscillates between $c/2$ and c (z between $c/8$ and $c/2$). With $H^2 = r_0^2 v_0^2 = c^3 g/4$, $E/m = \tfrac58 gc$. For $c = 1$ m: $\Omega_0 = v_0 = 3.132$ (rad/s and m/s), $\omega_r(r_0) = \tfrac{4}{\sqrt5}\Omega_0 = 5.603$ rad/s for the small oscillations of (e), which do not apply here because the amplitude is large.

At the turning points $\dot r = 0$, so $q^2\ddot r = \dfrac{H^2}{r^3} - \dfrac{gr}{c}$:

- $r = c/2$: $\dfrac{H^2}{r^3} = \dfrac{c^3 g/4}{c^3/8} = 2g$, $\dfrac{gr}{c} = \dfrac{g}{2}$, $q^2 = \dfrac54$: $\ddot r = \dfrac{3g/2}{5/4} = \dfrac65 g$.
- $r = c$: $\dfrac{H^2}{r^3} = \dfrac{g}{4}$, $\dfrac{gr}{c} = g$, $q^2 = 2$: $\ddot r = -\dfrac{3g}{8}$.

Then $N = mq\Big(\dfrac{r\ddot r}{c} + g\Big)$: at $r = c/2$, $N = m\dfrac{\sqrt5}{2}\Big(\dfrac35 g + g\Big) = \dfrac{4}{\sqrt5}mg \approx 1.789\,mg$; at $r = c$, $N = m\sqrt2\Big(-\dfrac38 g + g\Big) = \dfrac{5\sqrt2}{8}mg \approx 0.884\,mg$.

Step 4. Contact along any motion

From Step 3 of (c), multiplied by r: $r\ddot r\,q^2 = \dfrac{H^2}{r^2} - \dfrac{r^2\dot r^2}{c^2} - \dfrac{gr^2}{c}$. Adding $q^2\dot r^2$ to both sides and using $q^2 - r^2/c^2 = 1$:

$$q^2\left(\dot r^2 + r\ddot r\right) = \dot r^2 + \frac{H^2}{r^2} - \frac{gr^2}{c}$$

From the energy of (c), $q^2\dot r^2 = 2E/m - H^2/r^2 - gr^2/c$. Multiplying the previous line by q^2 and substituting $q^2\dot r^2$:

$$q^4\left(\dot r^2 + r\ddot r\right) = \frac{2E}{m} - \frac{H^2}{r^2} - \frac{gr^2}{c} + q^2\frac{H^2}{r^2} - q^2\frac{gr^2}{c} = \frac{2E}{m} + \frac{H^2}{c^2} - \left(2 + \frac{r^2}{c^2}\right)\frac{gr^2}{c}$$

because $(q^2-1)\dfrac{H^2}{r^2} = \dfrac{H^2}{c^2}$ and $(1+q^2)\dfrac{gr^2}{c} = \left(2+\dfrac{r^2}{c^2}\right)\dfrac{gr^2}{c}$. So

$$\dot r^2 + r\ddot r = \frac{1}{q^4}\left[\frac{2E}{m} + \frac{H^2}{c^2} - \left(2 + \frac{r^2}{c^2}\right)\frac{gr^2}{c}\right]$$

Substituting in $N = mq\left[(\dot r^2 + r\ddot r)/c + g\right] = \dfrac{m}{cq^3}\left[q^4(\dot r^2 + r\ddot r) + gc\,q^4\right]$ and using the identity

$$gc\,q^4 - \left(2 + \frac{r^2}{c^2}\right)\frac{gr^2}{c} = gc\left(1 + \frac{2r^2}{c^2} + \frac{r^4}{c^4}\right) - \frac{2gr^2}{c} - \frac{gr^4}{c^3} = gc$$

we get $N = \dfrac{m}{cq^3}\left(\dfrac{2E}{m} + \dfrac{H^2}{c^2} + gc\right)$, and with $cq^3 = (c^2+r^2)^{3/2}/c^2$:

$$N = \frac{m\left(2c^2E/m + H^2 + gc^3\right)}{(c^2+r^2)^{3/2}}$$

With $V = mgz \geq 0$ the energy is positive, so $N > 0$ on **every** motion: the particle never leaves the inner face. In the numerical case $2c^2E/m + H^2 + gc^3 = \tfrac{5}{4}gc^3 + \tfrac{1}{4}gc^3 + gc^3 = \tfrac{5}{2}gc^3$, so $N = \dfrac{5mgc^3}{2(c^2+r^2)^{3/2}}$, decreasing from $1.789\,mg$ at $r = c/2$ to its minimum $0.884\,mg$ at $r = c$, in agreement with Step 3.

KEY RESULTS

- $(u - r_0^2)\left(\dfrac{gu}{c} - v_0^2\right) = 0$: turning radii r_0 and $r^* = v_0\sqrt{c/g} = v_0/\Omega_0$.
- Case: $r^* = c$; $\ddot r(c/2) = \frac{6}{5}g$ (outwards), $\ddot r(c) = -\frac{3}{8}g$; $N(c/2) = \frac{4}{\sqrt5}mg \approx 1.789\,mg$, $N(c) = \frac{5\sqrt2}{8}mg \approx 0.884\,mg$.
- On every motion $N = \dfrac{m\,(2Ec^2/m + H^2 + gc^3)}{(c^2+r^2)^{3/2}} > 0$: contact is never lost. Here $N = \dfrac{5mgc^3}{2(c^2+r^2)^{3/2}}$.

Interpretation and checks

- **Dimensional check:** c is a length, so g/c and $\omega_r^2 = 4gc/(c^2+r_0^2)$ are s^{-2}; $H = r^2\dot\varphi$ is $\mathrm{m}^2\ \mathrm{s}^{-1}$, H^2/r^3 and gr/c are $\mathrm{m}\ \mathrm{s}^{-2}$; in N, $(2c^2E/m + H^2 + gc^3)$ is $\mathrm{m}^4\ \mathrm{s}^{-2}$ and $(c^2+r^2)^{3/2}$ is m^3: N is $\mathrm{kg}\ \mathrm{m}\ \mathrm{s}^{-2}$.
- **Limits:** $c \to \infty$ (flat floor): $\Omega_0 \to 0$, no horizontal circle without a wall. $r_0 \ll c$: the bowl acts as an isotropic linear spring, $\omega_r = 2\Omega_0$ and the orbits are centred ellipses.
- **Common mistakes in the exam:** (1) claiming that the angular momentum about O is conserved because "the reaction is normal": only the axial component is; (2) expecting the orbital rate to depend on r_0 (it does not on a paraboloid); (3) linearising without the factor $1 + r_0^2/c^2$, which gives $\omega_r = 2\Omega_0$ for every radius; (4) solving the quartic for the turning points instead of noticing that r_0 is a root; (5) not checking $N \geq 0$ for a unilateral contact.